

98% w/w sulfuric acid @ 70°C

1) High Si Iron

5% aqueous sodium hydroxide @ 30°C

Cast Iron

10% w/w hydrofluoric acid

Aluminium Bronze

High Si Iron x

Cast Nickel

Nickel-Copper alloys ✓ cheapest, best option.

5% w/w sulfuric acid @ 30°C

Aluminium Bronze

Copper ✓ cheapest, best option.

High Si Iron

Lead

5% nitric acid @ 30°C

High Si Iron

2) Rate of Corrosion

316

$$\frac{(4.5 - 4.026)}{2 \cdot 2 \cdot 7} = 0.0169 \frac{\text{in}}{\text{year}} \rightarrow 7 \text{ years to corrode.}$$

3

T.S.S

$$= \frac{316}{100} = 0.0169 \frac{\text{in}}{\text{year}} \rightarrow 14.79 \text{ years to corrode.}$$

1

304

$$0.0169 \cdot 0.9 = 0.0152 \frac{\text{in}}{\text{year}} \rightarrow 7.78 \text{ years to corrode.}$$

2

Cost

unit length: $\frac{\text{in}}{\text{yr}}$

↑	TSS	\$28
	unit	
	304	\$26
	unit	
	316	\$24
	unit	

Last 15 yr

316

$$0.0169 \cdot 15 = 0.2535 \text{ in} \cdot \frac{\$24}{\text{in}} = \$6.08 + \$60 = \$66.08$$

304

$$0.0152 \cdot 15 = 0.228 \text{ in} \cdot \frac{\$26}{\text{in}} = \$5.93 + \$40 = \$45.93$$

TSS

$$0.0169 \cdot 15 = 0.2535 \text{ in} \cdot \frac{\$28}{\text{in}} = \$7.098 + \$20 = \$27.10$$

* Use the TSS